

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) Semester II Examination, 2013-14

Paper: BCS-201 — Digital Logic Circuits

Time: Three hours

Maximum Marks: 70

*Instructions: Answer any **five** questions, including Question 1, which is compulsory.***Question 1.**

- (a) Perform the following conversions. Show the detailed calculation being done for conversions: [3×2 = 6]
- (i) $(673.23)_{10} = ()_2$
- (ii) $(1101101.111)_2 = ()_{10}$
- (iii) $(2AC5.D)_{16} = ()_8$
- (b) Represent the decimal number 3620 in BCD and excess-3 code. [2]
- (c) What is race around condition? [2]
- (d) List the names of any four digital logic families. [2]
- (e) What is the advantage of PLA over ROM? [2]

Question 2.

- (a) Perform the following binary operations: [3]
- (i) 111×1101
- (ii) $11011001 \div 1100$
- (b) Subtract 3 from -12 using 2's complement method. [2]
- (c) Perform the addition of the following numbers in the given base without converting to decimal: [4]
- (i) $(1230)_4 + (23)_4$
- (ii) $(296)_{12} + (57)_{12}$
- (d) What do you understand by floating point representation? Represent $(4.21875)_{10}$ in binary floating point notation. [5]

Question 3.

- (a) Simplify the following Boolean expressions to a minimum number of literals by using Boolean algebra theorems: [4]

(i) $xyz - x'y - xyz'$

(ii) $BC + AC' + AB + BCD$

- (b) What is the difference between canonical form and standard form? Which form is preferable when implementing a Boolean function with gates? Which form is obtained when reading a function from a truth table? [5]

- (c) Express the following function in a sum of minterms and product of maxterms:

$$F(w, x, y, z) = y'z + wxy' + wxz' + w'x'z$$

[5]

Question 4.

- (a) What are the differences between combinational circuit and sequential circuit? Explain with their block diagrams. List the names of any two combinational and any two sequential circuits. [4]

- (b) Implement the following function with NAND gates (with no more than six gates, each having three inputs). Assume that both the normal and complement inputs are available.

$$F = BD + BCD + AB'C'D' + A'B'CD'$$

[5]

- (c) Simplify the Boolean function F using the don't care condition d , in its minimum sum of product form:

$$F(w, x, y, z) = w'x'y'z' + w'x'yz' + w'xy'z + w'xyz'$$

$$d(w, x, y, z) = w'x'y'z + yz' + wyz$$

[5]

Question 5.

- (a) Design a combinational circuit that accepts a three bit number and generates an output binary number equal to the square of the input number. [5]

- (b) What is a decoder? Design a full adder circuit with a decoder and two OR gates. [4]

- (c) What is meant by a priority encoder? How does a priority encoder differ from an ordinary encoder? Explain with suitable example. [5]

Question 6.

- (a) Define Multiplexer. Implement an 8x1 multiplexer using two 4x1 multiplexers and a 2x1 multiplexer. [5]

- (b) Implement a 3-bit binary to gray code converter using PLA as follows:

- (i) Make a truth table. [2]

- (ii) Simplify using K-map. [2]

(iii) Design circuit using PLA; and [3]

(iv) Specify the size of PLA. [2]

Question 7.

(a) Implement JK flip flop using D-flip flop. [4]

(b) Define irregular counter. Design a counter with the following binary sequence: 0, 1, 3, 7, 6, 4 and repeat. Use T flip-flops. [5]

(c) Define register. By using registers, design a sequential circuit whose state table is listed as follows: [5]

Present State		Input	Next State		Output
A_1	A_2	x	A_1	A_2	y
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	0	1	0
0	1	1	0	0	1
1	0	0	1	0	0
1	0	1	0	1	0
1	1	0	1	1	0
1	1	1	0	0	1